

Администрирование сетевых подсистем

Лабораторная работа №6 — Установка и настройка MariaDB

Талебу Тенке Франк Устон

Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Цель лабораторной работы

Овладеть практическими навыками установки, конфигурирования и администрирования системы управления базами данных MariaDB.

Выполнение лабораторной работы

Установка пакетов MariaDB

```
root@server:~  
[vagrant@server ~]$ sudo -i  
[root@server ~]# dnf -y install mariadb mariadb-server  
Last metadata expiration check: 2:13:44 ago on Mon 09 Mar 2026 06:25:46 AM UTC.  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repository	Size
---------	------	---------	------------	------

```
=====
```

Installing:

mariadb	x86_64	3:10.5.29-3.el9_7	appstream	1.6 M
mariadb-server	x86_64	3:10.5.29-3.el9_7	appstream	9.7 M

Upgrading:

selinux-policy	noarch	38.1.65-1.el9_7.1	baseos	37 k
selinux-policy-targeted	noarch	38.1.65-1.el9_7.1	baseos	6.5 M

Installing dependencies:

mariadb-common	x86_64	3:10.5.29-3.el9_7	appstream	27 k
mariadb-errmsg	x86_64	3:10.5.29-3.el9_7	appstream	210 k
mysql-selinux	noarch	1.0.14-1.el9_6	appstream	36 k
perl-DBD-MariaDB	x86_64	1.21-17.el9	appstream	149 k
perl-Sys-Hostname	x86_64	1.23-481.1.el9_6	appstream	15 k

Installing weak dependencies:

mariadb-backup	x86_64	3:10.5.29-3.el9_7	appstream	6.5 M
mariadb-gssapi-server	x86_64	3:10.5.29-3.el9_7	appstream	14 k
mariadb-server-utils	x86_64	3:10.5.29-3.el9_7	appstream	210 k

Transaction Summary

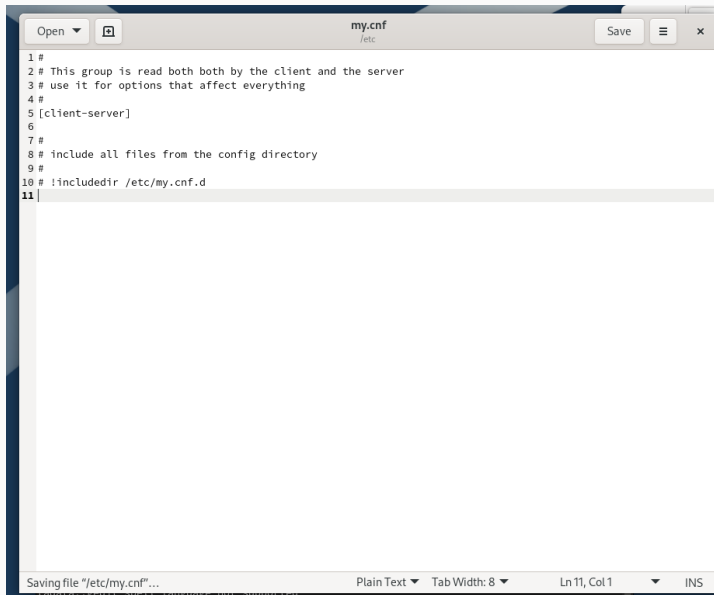
```
=====
```

Install 10 Packages
Upgrade 2 Packages

Total download size: 25 M
Downloading Packages:

(1/12): mariadb-common-10.5.29-3.el9_7.x86_64.rpm	87 kB/s 27 kB	00:00
(2/12): mariadb-errmsg-10.5.29-3.el9_7.x86_64.rpm	56 kB/s 210 kB	00:03
(3/12): mariadb-10.5.29-3.el9_7.x86_64.rpm	395 kB/s 1.6 MB	00:04
(4/12): mariadb-gssapi-server-10.5.29-3.el9_7.x86_64.rpm	76 kB/s 14 kB	00:00
(5/12): mariadb-server-utils-10.5.29-3.el9_7.x86_64.rpm	713 kB/s 210 kB	00:00
(6/12): mysql-selinux-1.0.14-1.el9_6.noarch.rpm	185 kB/s 36 kB	00:00
(7/12): perl-DBD-MariaDB-1.21-17.el9.x86_64.rpm	562 kB/s 149 kB	00:00
(8/12): mariadb-backup-10.5.29-3.el9_7.x86_64.rpm	1.2 MB/s 6.5 MB	00:05
(9/12): perl-Sys-Hostname-1.23-481.1.el9_6.x86_64.rpm	32 kB/s 15 kB	00:00
(10/12): selinux-policy-38.1.65-1.el9_7.1.noarch.rpm	56 kB/s 37 kB	00:00
(11/12): mariadb-server-10.5.29-3.el9_7.x86_64.rpm	3.5 MB/s 9.7 MB	00:02
(12/12): selinux-policy-targeted-38.1.65-1.el9_7.1.noarch.rpm	2.4 MB/s 6.5 MB	00:02

Настройка безопасности

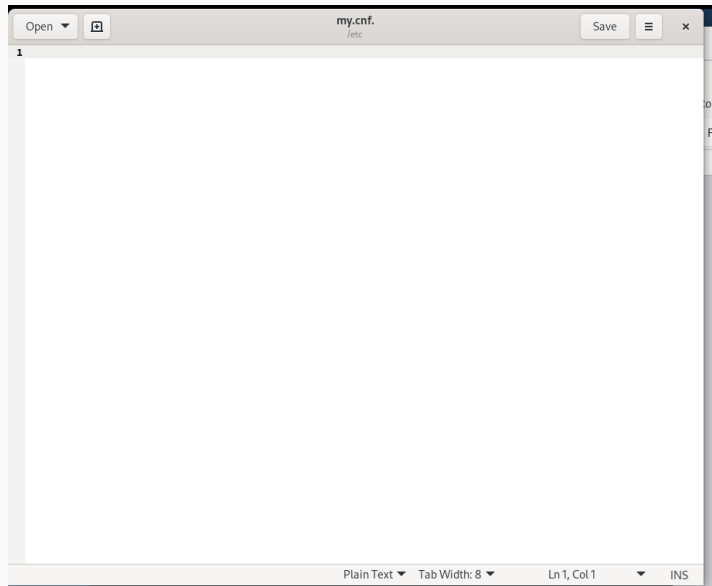


The screenshot shows a text editor window with a title bar that includes 'my.cnf' and '/etc'. The window has 'Open', 'Save', and a close button. The text content is as follows:

```
1 #
2 # This group is read both both by the client and the server
3 # use it for options that affect everything
4 #
5 [client-server]
6
7 #
8 # include all files from the config directory
9 #
10 # !includedir /etc/my.cnf.d
11 |
```

The status bar at the bottom indicates 'Saving file "/etc/my.cnf"...', 'Plain Text', 'Tab Width: 8', 'Ln 11, Col 1', and 'INS'.

Список клиентских команд



Просмотр доступных баз данных

```
** (gedit:18155): WARNING **: 08:48:46.983: Set document metadata failed: Setting attribute metadata
::gedit-position not supported

(gedit:18155): dconf-WARNING **: 08:48:47.002: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)
[root@server etc]# systemctl start mariadb
[root@server etc]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/syste
m/mariadb.service.
[root@server etc]# ss -tulpen | grep mysql
[root@server etc]# netstat -tulpen | grep 3306
bash: prep-w: command not found...
[root@server etc]# netstat -tulpen | grep 3306
[root@server etc]# netstat -tulpen | grep 3306
tcp6      0      0 :::3306          :::*              LISTEN     27          781624
18288/mariadb
[root@server etc]#
```

Рис. 4: SHOW DATABASES

Статус MariaDB до изменения кодировки

```
** (gedit:18155): WARNING **: 08:48:46.983: Set document metadata failed: Setting attribute metadata
::gedit-position not supported

(gedit:18155): dconf-WARNING **: 08:48:47.002: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)
[root@server etc]# systemctl start mariadb
[root@server etc]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/syste
m/mariadb.service.
[root@server etc]# ss -tulpen | grep mysql
[root@server etc]# netstat -tulpen | grep 3306
bash: prep-w: command not found...
[root@server etc]# netstat -tulpen | grep 3306
[root@server etc]# netstat -tulpen | grep 3306
tcp6      0      0 :::3306                :::*                    LISTEN      27          781624
        18288/mariabdb
[root@server etc]#
```

Рис. 5: Статус MariaDB — до настройки UTF-8

Создание файла utf8.cnf

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] Y
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] Y
New password:
Re-enter new password:
Sorry, you can't use an empty password here.

New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n]

Examine

Статус MariaDB после настройки UTF-8

```
root@server:/etc

Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] Y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] T
Disallow root login remotely? [Y/n] Y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] Y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] Y
... Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
[root@server etc]#
```

Low Disk Space
This computer has only 4

Examine

Создание таблицы и добавление данных

Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> \h

General information about MariaDB can be found at
<http://mariadb.org>

List of all client commands:

Note that all text commands must be first on line and end with ';'.

?	(\?)	Synonym for 'help'.
charset	(\C)	Switch to another charset. Might be needed for processing binlog with multi-byte char sets.
clear	(\c)	Clear the current input statement.
connect	(\r)	Reconnect to the server. Optional arguments are db and host.
delimiter	(\d)	Set statement delimiter.
edit	(\e)	Edit command with \$EDITOR.
ego	(\G)	Send command to MariaDB server, display result vertically.
exit	(\q)	Exit mysql. Same as quit.
go	(\g)	Send command to MariaDB server.
help	(\h)	Display this help.
nopager	(\n)	Disable pager, print to stdout.
notee	(\t)	Don't write into outfile.
nowarning	(\w)	Don't show warnings after every statement.
pager	(\P)	Set PAGER [to_pager]. Print the query results via PAGER.
print	(\p)	Print current command.
prompt	(\R)	Change your mysql prompt.
quit	(\q)	Quit mysql.
rehash	(\#)	Rebuild completion hash.
sandbox	(\-)	Disallow commands that access the file system (except \P without an argument and \e).
source	(\.)	Execute an SQL script file. Takes a file name as an argument.
status	(\s)	Get status information from the server.
system	(\!)	Execute a system shell command.
tee	(\T)	Set outfile [to_outfile]. Append everything into given outfile.
use	(\u)	Use another database. Takes database name as argument.
warnings	(\W)	Show warnings after every statement.

For server side help, type 'help contents'

MariaDB [(none)]>

Examine

Создание пользователя и назначение привилегий

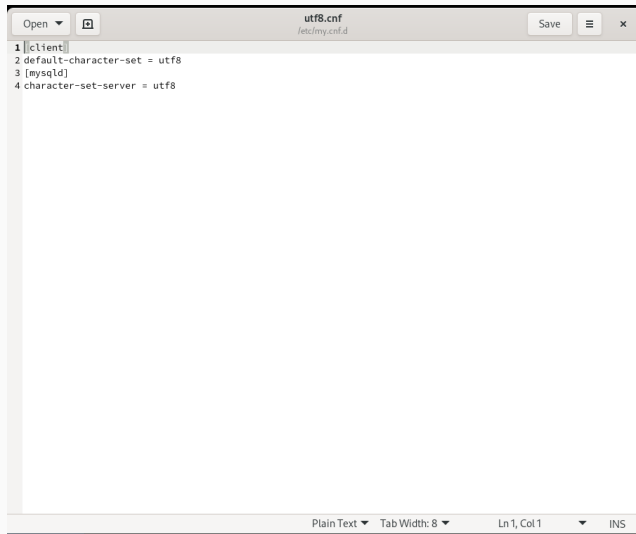
```
For server side help, type 'help contents'

MariaDB [(none)]> SHOW DATABASES
-> SHOW DATABASES;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near 'SHOW DATABASES' at line 2
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
+-----+
3 rows in set (0.014 sec)

MariaDB [(none)]> 
```

Рис. 9: Права пользователя

Просмотр структуры таблицы



Просмотр списка таблиц

```
(gedit:18481): dconf-WARNING **: 09:10:12.511: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)
[root@server my.cnf.d]# systemctl restart mariadb
[root@server my.cnf.d]# gedit utf8.cnf

(gedit:18617): dconf-WARNING **: 09:10:55.372: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)

(gedit:18617): dconf-WARNING **: 09:10:55.376: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)

(gedit:18617): dconf-WARNING **: 09:10:55.578: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)

(gedit:18617): dconf-WARNING **: 09:10:55.579: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)

(gedit:18617): dconf-WARNING **: 09:10:55.580: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)

** (gedit:18617): WARNING **: 09:11:08.717: Set document metadata failed: Setting attribute metadata
::gedit-position not supported

(gedit:18617): dconf-WARNING **: 09:11:08.766: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)
[root@server my.cnf.d]# systemctl restart mariadb
[root@server my.cnf.d]# mysql -u root -p
Enter password:
```

Рис. 11: mysqlshow addressbook

Создание резервных копий

```
(gedit:18617): dconf-WARNING **: 09:11:08.766: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)
[root@server my.cnf.d]# systemctl restart mariadb
[root@server my.cnf.d]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          3
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.5.29-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    latin1
Db characterset:        latin1
Client characterset:    utf8
Conn. characterset:     utf8
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 12 min 24 sec

Threads: 1  Questions: 4  Slow queries: 0  Opens: 17  Open tables: 10  Queries per second avg: 0.005
-----

MariaDB [(none)]>
```


Скрипт автоматической настройки MariaDB

```
root@server:/etc/my.cnf.d

SSL:                               Not in use
Current pager:                     stdout
Using outfile:                     ''
Using delimiter:                   ;
Server:                             MariaDB
Server version:                    10.5.29-MariaDB MariaDB Server
Protocol version:                  10
Connection:                        Localhost via UNIX socket
Server characterset:               latin1
Db characterset:                   latin1
Client characterset:               utf8
Conn. characterset:                utf8
UNIX socket:                       /var/lib/mysql/mysql.sock
Uptime:                            12 min 24 sec

Threads: 1  Questions: 4  Slow queries: 0  Opens: 17  Open tables: 10  Queries per second avg: 0.005
-----

MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> USE addressbook;
Database changed
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.003 sec)

MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.097 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов','Москва');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near 'INSERT INTO city(name,city) VALUES ('Иаанов',
'Москва')' at line 1
MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов','Москва');
Query OK, 1 row affected (0.026 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Петров','Сочи');
Query OK, 1 row affected (0.014 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Сидоров','Дубна');
Query OK, 1 row affected (0.013 sec)

MariaDB [addressbook]>
```

Выводы

Итоги выполненной лабораторной работы

В результате работы была проведена полная установка и настройка MariaDB:

- Установлены серверные и клиентские пакеты.
- Настроена система безопасности MariaDB.
- Созданы база данных и таблицы, добавлены данные.
- Настроены пользователи и привилегии.
- Созданы резервные копии и выполнено восстановление базы.
- Реализована автоматизация развёртывания с помощью скрипта `mysql.sh`.

Получены навыки администрирования серверов баз данных и управления окружением MariaDB.